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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Francium fluoride
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG track 4: Quality education
Project / Solution Name	Future Skill AI
Name (Leader)	Chew Yu Hang
Email & Phone (Leader)	zhouyuxing931@gmail.com & +60 16-246 7036
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In a rapidly changing job market shaped by artificial intelligence and emerging technologies, students, graduates, and career changers face increasing difficulty identifying suitable career paths and preparing for the skills those careers require. Career information, labour-market trends, educational resources, and sustainability initiatives are often scattered across different platforms, making it challenging to turn career interests into realistic, actionable learning plans.

This problem particularly affects young people entering the workforce and individuals with limited access to personalised career guidance. Without a clear understanding of their strengths, existing competencies, and skill gaps, they may struggle to select appropriate career paths, invest their time in relevant training, or adapt to evolving employment demands. These difficulties can contribute to skills mismatches, underemployment, and unequal access to career development opportunities.

Furthermore, individuals may overlook how their chosen occupations can contribute to broader societal goals, including quality education (SDG 4), decent work & economic growth (SDG 8), and sustainable development. Career planning should therefore extend beyond finding a job; it should help individuals understand their potential, adapt to change, and recognize the wider impact of their work.

Our team aims to address this challenge by making career exploration, skill development, and sustainability awareness more accessible through an AI assisted platform that transforms career aspirations into personalised, practical reskilling pathways.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FutureSkill AI is an AI-assisted career development platform that connects career discovery, skills analysis, personalised learning, and sustainable development in one accessible experience.

Users can explore AI-driven career trends to understand how emerging technologies may reshape employment opportunities and use the “Employability Predictor” to investigate uncertainty surrounding future career prospects. A career assessment helps users identify careers aligned with their interests and strengths. Through personalised Sustainable Development Goal (SDG) mapping, users can also understand how different occupations and their required skills may contribute to societal priorities, including SDG 4: Quality Education and SDG 8: Decent Work and Economic Growth.

Beyond career discovery, users can upload their CVs, specify a target career, provide their existing skills, and indicate how much time they can dedicate to learning. FutureSkill AI analyses the gaps between their current capabilities and the skills required for their chosen career, generating a personalised reskilling roadmap organised around practical learning tasks and available time. Generative AI helps translate identified skill gaps into actionable guidance, while the career-matching system provides the foundation for recommendations.

For users seeking further preparation, an AI-powered interview practice interface is designed to simulate an interviewer, ask career-relevant questions, and help users improve their job readiness. Its current functionality depends on resolving an external AI API-key configuration. To encourage sustained engagement, a reward system awards experience points for completed learning tasks, allowing users to grow a digital tree that visualises their learning progress.

The project leverages skills developed through the SDC Phase 1 workshops, including generative AI integration, API development, database architecture, and cloud hosting. We use Python and FastAPI for backend services, Replit for application development and deployment, and GitHub for version control. By combining structured career-matching logic with generative AI, FutureSkill AI aims to deliver personalised career guidance and practical reskilling pathways grounded in user-provided information and defined skill requirements.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

FutureSkill AI uses Python and FastAPI for backend development, with HTML, CSS and JavaScript for the frontend. Its career-matching engine processes assessment responses and structured career profiles stored in CSV files to recommend suitable careers and identify skill gaps. The application is deployed on Render, while GitHub supports source-code version control.

The system connects the frontend to the FastAPI backend, which processes user inputs and invokes the career-matching engine. The resulting career recommendations and skill-gap information are returned to the front end. Generative AI is intended to generate personalised reskilling roadmaps and support interview practice. However, the external AI API integration for the interview webpage is currently non-functional, so this feature is not yet operational. We have not used RunPod GPU infrastructure to host a model.

Our current prototype uses CSV files for structured career data; a persistent database for user profiles and learning progress remains a potential future enhancement. The architecture combines structured career-matching logic with planned generative AI integration, while Render provides application hosting and GitHub manages the codebase.

Regarding SDC sponsor-provided technology, we have not deployed a model using RunPod GPU resources or confirmed the integration of a sponsor-provided GenAI API. We therefore distinguish our actual implementation from planned AI capabilities and do not claim to have used sponsor-provided inference infrastructure.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We built FutureSkill AI around two questions: “What careers could suit me?” and “What should I learn next?” Our design-thinking process focused on students and early-career learners who need clear, achievable steps.

We defined the journey as discover, explore and grow, then prototyped a connected flow from the 42-question assessment and 923-career catalogue to skill-gap analysis and learning pathways for supported roles. Iterative reviews helped us improve mobile layouts, light/dark readability and SDG evidence navigation. Users can review their inputs, browse careers independently and track completed learning activities.

For technical viability, we integrated the modules into one Python application with SQLite persistence and a GitHub-to-Render deployment workflow. All 93 Python tests passed, alongside browser checks covering core journeys and language switching. Career and SDG outputs remain linked to their inputs and available evidence.

By Phase 2, we will pilot the platform with target users, strengthen authentication and upload security, and introduce deployment monitoring. Pilot findings will guide our priorities, while repeatable release checks will help us move confidently from a working prototype towards wider deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4 " By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship"
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Baseline: 0% of our pilot users have yet been measured as successfully completing a personalised, AI-assisted skill-gap analysis and reskilling roadmap through FutureSkill AI. We will establish the actual baseline through initial prototype user testing.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% of test users who successfully generate and understand a personalized skill-gap analysis and reskilling roadmap aligned with their target career and available learning time.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve at least an 80% successful task-completion rate among pilot testers by the final prototype presentation, measured through structured user testing and feedback.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✅ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✅ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI and Pydantic	Code — API framework and data validation	MIT
Uvicorn, Starlette and HTTPX	Code — application server, web components and HTTP client	BSD-3-Clause
Flask, Werkzeug, Jinja2 and itsdangerous	Code — interview module and supporting web libraries	BSD-3-Clause
pandas and NumPy	Code — data processing and numerical calculations	BSD-3-Clause; additional bundled-component notices apply
python-multipart and a2wsgi	Code — file-upload parsing and application integration	Apache-2.0
OpenAI Python SDK	Code — access to hosted AI services	Apache-2.0

Apache ECharts	Code — interactive charts	Apache-2.0
SQLite	Code — local database	Public domain
DM Sans and Space Mono	Code/assets — website fonts	SIL Open Font License 1.1
O*NET Database 31.0 — Task Statements	Dataset — occupational tasks used for SDG analysis	CC BY 4.0; attribution to USDOL/ETA
Imported 923-career profile CSV	Dataset — occupational titles and interest-profile values	Original source and licence require verification
United Nations SDG definitions and official Goal icons	Reference material and visual assets	Subject to UN copyright and SDG communications usage guidelines
Pexels photographs	Visual assets — education and workplace imagery	Pexels License
OpenAI GPT-4.1 mini	Model — optional explanations and interview generation/feedback	Proprietary, accessed under applicable OpenAI API/service terms
OpenAI text-embedding-3-small	Model — optional interview-question similarity checks	Proprietary, accessed under applicable OpenAI API/service terms
OpenAI gpt-4o-mini-transcribe	Model — optional audio transcription	Proprietary, accessed under applicable OpenAI API/service terms
World Economic Forum, *Future of Jobs Report 2025*	Reference publication cited by the trends module	WEF copyright and applicable reuse terms; not an openly licensed dataset

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✅ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10/2026